

Common Structure of SOPs

A common philosophy has been established as the foundation for each SOP, ensuring consistent content throughout the Process.

All mechanical components are classified and defined as **Critical Functional Elements**, whose status directly determines the health of the Process.

1. General Description and Function

Contains: A technical description of the equipment or process, its main function within the line, and its interrelationship with the preceding and subsequent stages.

It provides the essential context so that, regardless of experience level, anyone can understand **the "why" and the "what for"** of each machine within the overall process, enabling a comprehensive overview and facilitating informed decision-making.

2. Critical Points and Risk Management

Contains: An identification of the mechanical or electrical components within the process stages that require maximum attention and control to prevent machine failures that could affect equipment efficiency or tile quality, along with some established measures to mitigate these effects.

Most of these components are classified in the Manuals as **WP (Wear Part) or CP (Critical Part)**. This approach allows efforts to be concentrated on the most sensitive aspects and act proactively, before a problem materializes into a breakdown or repair.

3. SUPERVISION LEVEL – DAILY

Contains: A list of visual, auditory, manual, and basic checks that should be performed at the start or during each shift.

A list of the most significant components in the process and the assigned control level for each one is attached.

The objective is to carry out constant operational monitoring that **facilitates the early detection of anomalies** by the operator or assigned maintenance personnel, turning maintenance into a daily and shared task.

4. CLEANING / CHECK:

Contains: Specific instructions for the cleaning and basic inspection of key components of each component or equipment.

I believe that clean equipment is inspected equipment.

Systematic cleaning provides an **opportunity to identify wear, leaks, or dirt that affects the process.**

5. PREVENTIVE MAINTENANCE TASK

Contains: A schedule of possible maintenance tasks with an initial frequency (weekly, monthly, quarterly, annual) and step-by-step protocols for the replacement of wear components.

Planning is proposed for ongoing maintenance tasks, with replacement protocols **activated based on wear measurements and the control of each component.**

6. Maintenance Guidelines

(Lubrication, Wear Monitoring, Inspection Protocol)

*Detailed lubrication instructions.

*Methods for measuring wear.

*Frequency of in-depth inspections.

*Visual, sensory, and manual maintenance.

*Data-based actions and controls.

It combines:

- Prescriptive actions (e.g., do not lubricate the PTFE chain) with technical knowledge.
- Measurable indicators (spacing, length of multi-strand cords) to objectify wear.
- Scheduled reviews (monthly chain inspection) to verify condition.

7. Monitoring & Measurements

Contains: Instructions for taking specific measurements and maintaining a historical record of them.

It **is the central core of component and equipment monitoring and control**.

Creating a history of measurements provides objective reference data necessary for:

- Accurate diagnosis.
- Predicting failures.
- Justifying and planning modifications or investments in a controlled manner.

9. SHEET PROCESS - Reference Settings

Contains: The obligation to keep an updated sheet with all master reference settings for each product or format.

Process repeatability and traceability must be ensured.

These values are **the "base recipe" that guarantees quality and efficiency**.

Always starting from a known and recorded setting is essential for making any controlled modification and measuring its real impact.

Conclusion

Each section of the SOPs is not an isolated element, but a gear in a single system designed to:

1. **Transmit clear knowledge** to all levels.
2. **Anticipate and manage risks** (quality, safety, shutdown).
3. **Turn maintenance** into a daily, sensory, measurable activity adapted to real possibilities.
4. **Make decisions based on historical data** and objective reference points.
5. Enable continuous improvement through controlled and safe modifications.

This coherent structure ensures that, regardless of the specific machine, the approach to operating, maintaining, and improving it is always the same: **systematic, safe, data-driven, and focused on the final efficiency and quality of the tile**.